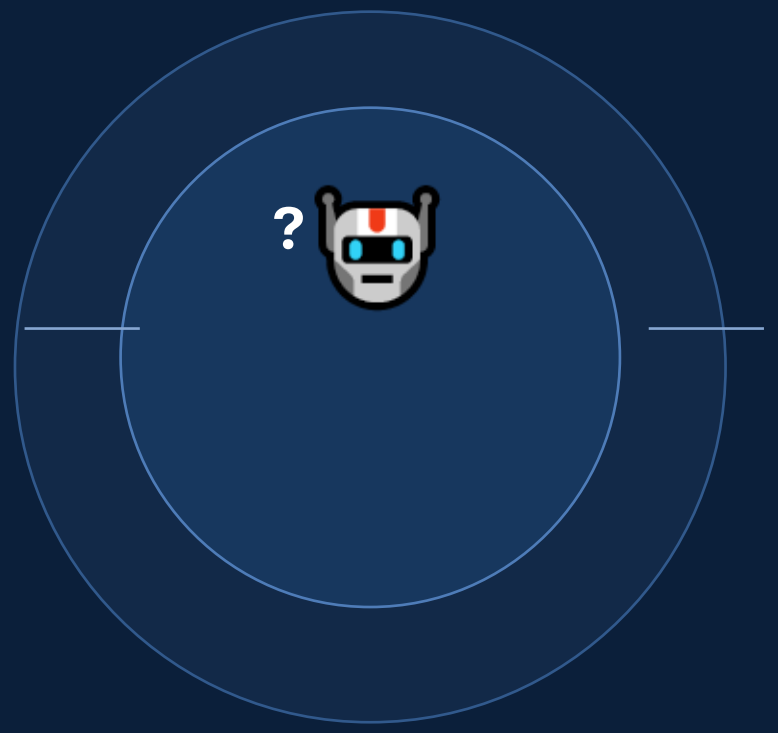
Search under Uncertainty

Nondeterministic Actions and Partial Observability



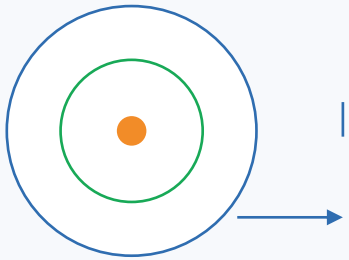
Primary source: Russell & Norvig, Artificial Intelligence: A Modern Approach, 4th ed.

Learning outcome

What students should know and be able to do

By the end of this lesson, students should be able to:

- Differentiate deterministic, nondeterministic, and partially observable search problems.
- Explain why uncertainty changes a solution from a fixed path to a conditional plan.
- Apply AND–OR search to problems with nondeterministic actions.
- Use belief states to represent uncertainty about the current state.
- Select an appropriate search strategy and justify modeling assumptions.



Core skill:
Choose the right search model
for the uncertainty you face.

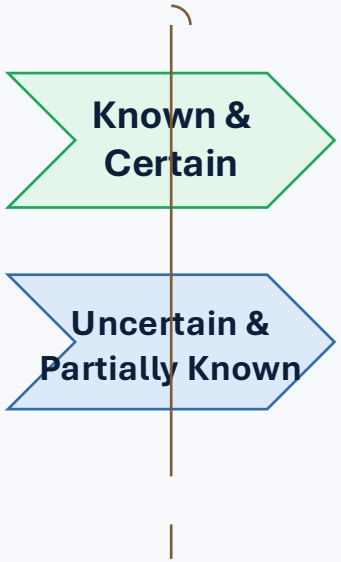
Why classical search is not enough

Classical search assumes a known state and predictable action outcomes.

Classical search assumptions

- Fully observable environment
- Deterministic actions
- Static, known world model
- Single known current state
- Solution is a fixed sequence of actions

Limitation: real environments often have uncertain action outcomes or incomplete information. The agent needs a search strategy that can reason about uncertainty.



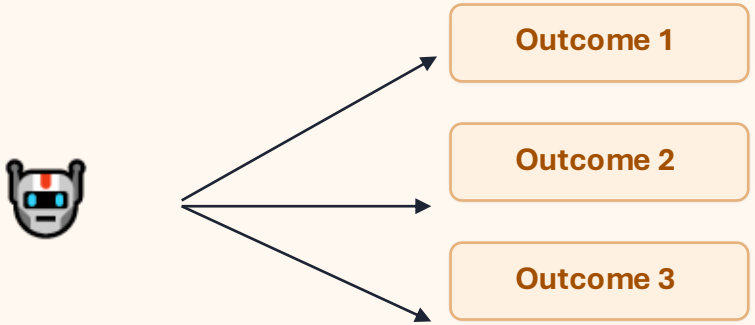
When certainty breaks, path search becomes search under uncertainty.

Two forms of uncertainty

These are related but not the same problem.

A. Nondeterministic actions

State is known, but one action may produce several outcomes.



B. Partial observability

Actions may be reliable, but the exact current state is unknown.

Search with nondeterministic actions

When actions have multiple possible results, the plan must account for all of them.

Definition

The same action in the same state can lead to several successor states.

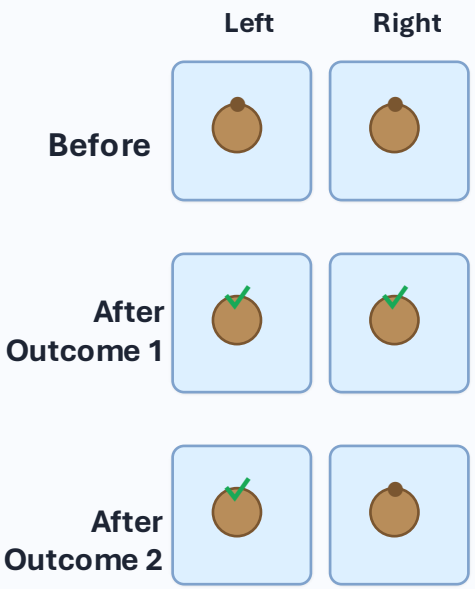
Design implication

A good plan still succeeds no matter which modeled outcome occurs.

Example: the erratic vacuum world

In the standard vacuum world, Suck reliably cleans the current square. In the erratic version, the same action may clean the current square, also clean an adjacent square, or fail to clean anything. Same action, different outcomes.

Teaching point: if one action can yield multiple successors, ordinary path search is insufficient.

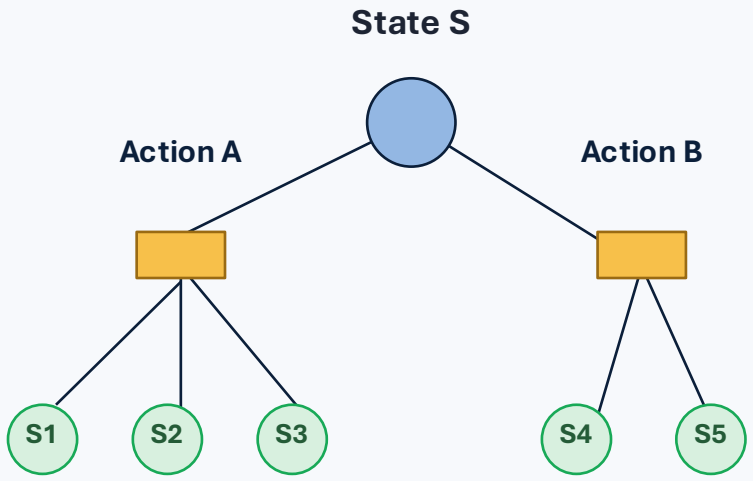


AND–OR search trees

OR = choose an action. AND = handle every possible outcome of that action.

Legend

- OR node (agent chooses)
- AND node (outcomes)



A solution is not a single path. It is a contingent plan that chooses an action at each OR node and specifies what to do for every outcome at each AND node.

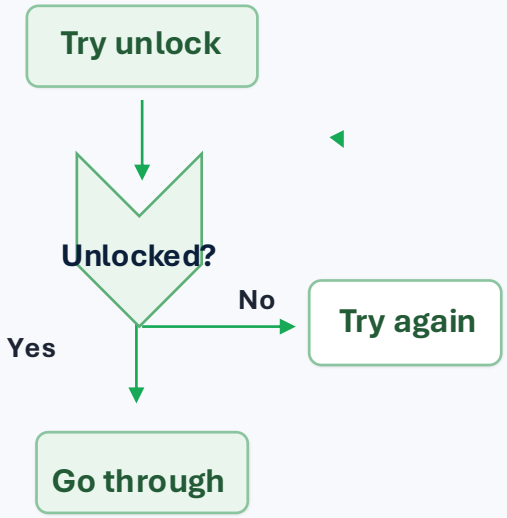
OR = choose one good action | AND = be prepared for every possible result

Cyclic plans and modeling assumptions

Some uncertain problems are solved by retrying; your assumptions determine whether that is sound.

“Try, try again”

A good plan may be cyclic. If an action can fail but can be tried again, the right solution may be a looping strategy rather than a finite action sequence.



Modeling assumptions to justify

- Are all possible action outcomes known?
- Can the agent safely retry, or does failure make the problem worse?
- Must the plan succeed for all outcomes, or only likely outcomes?
- Are illegal actions harmless, impossible, or dangerous?
- Is the environment merely uncertain, or adversarial?

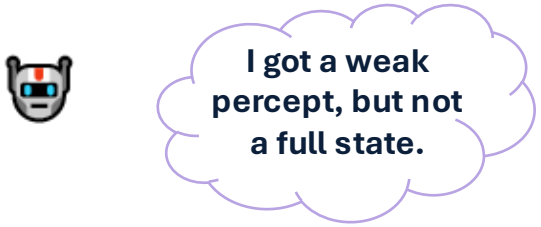
These assumptions are what make one search strategy defensible and another inappropriate.

Partial observability and belief states

Now the uncertainty is about where the agent is, not just what an action will do.

What is partial observability?

Sensors do not reveal the exact current state, so the agent reasons over a set of possible states.



Physical state

The actual world configuration.

A	B
C	D

One of these is the real state.

Belief state

The set of states the agent thinks it might be in.

{A, B}

"I know I am in A or B, but I am not sure which."

Belief-state search: prediction, correction, and legality assumptions

The search node is now a set of possible states, updated by actions and observations.

1. Prediction

From belief state b, apply action a and collect every state that could result.

$$b' = \text{Result}(b, a)$$

This is the sensorless case: only action effects update what is possible.

2. Correction / filtering

After a percept arrives, remove states inconsistent with that observation.

$$b'' = \{ s \in b' \mid \text{percept}(s) \text{ matches} \}$$

b double prime is the subset of b prime containing all states s whose percept matches

With partial observations, beliefs are updated using both actions and percepts.

Action legality assumptions

If a belief state contains many possible states, which actions are legal?

Union of legal actions

Safe only if illegal actions have no harmful effect.

Intersection of legal actions

Safer when taking an illegal action could be costly or dangerous.

Choosing the right search strategy

Match the method to the kind of uncertainty, then defend your assumptions.

Feature	Deterministic & Fully Observable	Nondeterministic & State Known	Partially Observable	Best fit
Current state known?	Yes	Yes	No, uncertain	—
Action outcome	One result	Multiple results	Often one result (but state hidden)	—
Solution form	Fixed sequence	Conditional plan / policy	Plan over belief states	—
Recommended approach	Classical path search	AND–OR search	Belief-state search	**Pick the simplest model that matches the uncertainty**

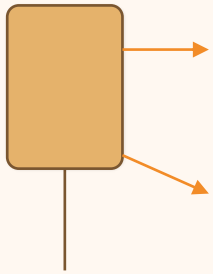
Justification language to use: "I selected AND–OR search because the current state is known but each action has several possible outcomes." / "I selected belief-state search because the exact current state is not directly observable."

Worked examples

Practice identifying the uncertainty and selecting the search strategy.

Nondeterministic example

A door may open or jam. The current room is known. Plan with AND–OR search so both outcomes are covered.



Partial observability example

A robot may be in the left or right room and receives only a weak “near wall” percept. Plan in belief-state space.

Mixed discussion prompt

If actions are nondeterministic and the state is only partially observed, what additional machinery beyond basic search might be needed?

Key takeaways

Select the search representation that matches the uncertainty present.

Key takeaways

- Deterministic search solves for fixed paths.
- Nondeterministic search solves for conditional plans, often with AND–OR trees.
- Partial observability requires belief states and search over sets of possible states.
- Prediction uses action effects; correction uses observations.
- The right strategy depends on justified assumptions about outcomes, observability, and action legality.